

Derivation of Wall Vorticity Boundary Conditions

Streamfunction–Vorticity Formulation for Lid-Driven Cavity

Computational Fluid Dynamics Technical Note

August 3, 2026

Abstract

In the streamfunction–vorticity (ψ – ω) formulation of the two-dimensional incompressible Navier–Stokes equations, boundary conditions for vorticity are not naturally prescribed on solid boundaries. This document presents a rigorous step-by-step mathematical derivation of wall vorticity boundary conditions (Thom’s formula) using Taylor series expansions. Specific formulas are detailed for all four walls of a lid-driven cavity flow domain and validated against standard numerical discretization scripts.

1 Detailed Derivation of Wall Vorticity Boundary Conditions

1.1 Introduction

In the streamfunction–vorticity formulation, the governing equations for 2D viscous incompressible flow are:

$$\frac{\partial \omega}{\partial t} + u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} = \nu \left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right) \quad (1)$$

and the kinematic Poisson equation:

$$\nabla^2 \psi = -\omega \quad (2)$$

The streamfunction ψ defines the velocity field through:

$$u = \frac{\partial \psi}{\partial y}, \quad \text{and} \quad v = -\frac{\partial \psi}{\partial x} \quad (3)$$

For a lid-driven cavity, the streamfunction on all domain walls is known (constant along solid walls). However, the wall vorticity is not directly known. Therefore, we require an explicit algebraic relation that connects:

- Wall streamfunction (ψ_w)
- Wall tangential velocity (U_t)
- Adjacent interior streamfunction (ψ_p)
- Wall vorticity (ω_w)

This relation is obtained systematically using a Taylor series expansion about the boundary.

1.2 Step 1: Local Coordinate System

Consider a general solid wall. Introduce two local orthogonal coordinates:

- s : tangential direction along the wall
- n : inward normal direction perpendicular to the wall

The Laplacian operator in local coordinates is written as:

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial s^2} + \frac{\partial^2 \psi}{\partial n^2} \quad (4)$$

Substituting this into the Poisson equation gives:

$$\frac{\partial^2 \psi}{\partial s^2} + \frac{\partial^2 \psi}{\partial n^2} = -\omega \quad (5)$$

1.3 Step 2: Streamfunction Behaviour on a Wall

An impermeable wall cannot be crossed by fluid, meaning the normal velocity component vanishes ($v_n = 0$). Consequently, the wall constitutes a streamline. Since the streamfunction remains constant along any streamline:

$$\psi = \text{constant along the wall} \quad (6)$$

Without loss of generality, we set:

$$\psi_{\text{wall}} = 0 \quad (7)$$

Because the value of ψ does not change along the wall (s -direction):

$$\frac{\partial \psi}{\partial s} = 0 \quad \implies \quad \frac{\partial^2 \psi}{\partial s^2} = 0 \quad (8)$$

This is a crucial physical and mathematical simplification.

1.4 Step 3: Reduction of the Poisson Equation at the Wall

Substituting $\frac{\partial^2 \psi}{\partial s^2} = 0$ into the local Poisson equation:

$$0 + \frac{\partial^2 \psi}{\partial n^2} = -\omega \quad (9)$$

evaluated directly at the wall yields:

$$\omega_{\text{wall}} = - \left(\frac{\partial^2 \psi}{\partial n^2} \right)_{\text{wall}} \quad (10)$$

This fundamental equation directly links the wall vorticity to the second normal derivative of the streamfunction at the boundary.

1.5 Step 4: Taylor Expansion

Consider a wall boundary node and the first interior node located at a normal distance h into the fluid domain. Let:

- ψ_w = wall streamfunction value
- ψ_p = first interior streamfunction value
- h = normal grid spacing perpendicular to the wall

Expanding ψ in a Taylor series about the wall node along the normal direction n :

$$\psi_p = \psi_w + \left(\frac{\partial \psi}{\partial n} \right)_w h + \left(\frac{\partial^2 \psi}{\partial n^2} \right)_w \frac{h^2}{2} + \mathcal{O}(h^3) \quad (11)$$

1.6 Step 5: Replace the First Derivative

The first normal derivative of the streamfunction corresponds to the tangential fluid velocity at the wall boundary:

$$\left(\frac{\partial \psi}{\partial n} \right)_w = U_t \quad (12)$$

where U_t is the prescribed tangential velocity of the wall. Substituting this into the Taylor expansion gives:

$$\psi_p = \psi_w + U_t h + \left(\frac{\partial^2 \psi}{\partial n^2} \right)_w \frac{h^2}{2} + \mathcal{O}(h^3) \quad (13)$$

1.7 Step 6: Replace the Second Derivative

Using the wall Poisson identity $\left(\frac{\partial^2 \psi}{\partial n^2} \right)_w = -\omega_w$:

$$\psi_p = \psi_w + U_t h - \omega_w \frac{h^2}{2} + \mathcal{O}(h^3) \quad (14)$$

1.8 Step 7: Solve for Wall Vorticity

Rearranging the equation to isolate the wall vorticity ω_w :

$$\psi_p - \psi_w - U_t h = -\omega_w \frac{h^2}{2} \quad (15)$$

Multiplying by $-\frac{2}{h^2}$ yields the classic first-order Thom boundary formula for wall vorticity:

$$\omega_w = \frac{2(\psi_w - \psi_p)}{h^2} + \frac{2U_t}{h} \quad (16)$$

2 Boundary Formulations for Lid-Driven Cavity Walls

We now apply the general formula $\omega_w = \frac{2(\psi_w - \psi_p)}{h^2} + \frac{2U_t}{h}$ to all four boundaries of a standard rectangular cavity discretized on a 2D Cartesian grid (i, j) with grid spacings dx and dy .

2.1 Bottom Wall ($j = 0$)

At the bottom boundary $j = 0$, the first interior point is $j = 1$, and $h = dy$. The wall is stationary, so $U_t = 0$:

$$\omega_{i,0} = \frac{2(\psi_{\text{wall}} - \psi_{i,1})}{dy^2} \quad (17)$$

Since $\psi_{\text{wall}} = 0$:

$$\omega_{i,0} = -\frac{2\psi_{i,1}}{dy^2} \quad (18)$$

2.2 Top Moving Lid ($j = ny - 1$)

At the top lid $j = ny - 1$, the first interior point is $j = ny - 2$, and $h = dy$. The inward normal points downwards ($-\hat{y}$), so the relative tangential velocity is $U_t = -U_{\text{lid}}$:

$$\omega_{i,ny-1} = \frac{2(\psi_{\text{wall}} - \psi_{i,ny-2})}{dy^2} - \frac{2U_{\text{lid}}}{dy} \quad (19)$$

With $\psi_{\text{wall}} = 0$:

$$\omega_{i,ny-1} = -\frac{2\psi_{i,ny-2}}{dy^2} - \frac{2U_{\text{lid}}}{dy} \quad (20)$$

2.3 Left Wall ($i = 0$)

At the left boundary $i = 0$, the first interior point is $i = 1$, and $h = dx$. The wall is stationary ($U_t = 0$):

$$\omega_{0,j} = \frac{2(\psi_{\text{wall}} - \psi_{1,j})}{dx^2} \quad (21)$$

With $\psi_{\text{wall}} = 0$:

$$\omega_{0,j} = -\frac{2\psi_{1,j}}{dx^2} \quad (22)$$

2.4 Right Wall ($i = nx - 1$)

At the right boundary $i = nx - 1$, the first interior point is $i = nx - 2$, and $h = dx$. The wall is stationary ($U_t = 0$):

$$\omega_{nx-1,j} = \frac{2(\psi_{\text{wall}} - \psi_{nx-2,j})}{dx^2} \quad (23)$$

With $\psi_{\text{wall}} = 0$:

$$\omega_{nx-1,j} = -\frac{2\psi_{nx-2,j}}{dx^2} \quad (24)$$

3 Verification with Python Implementation

The analytical formulas derived above directly map to the numerical boundary condition update step in finite-difference Navier–Stokes solvers:

```

1 # Bottom stationary wall (j = 0)
2 omega0[0, i] = 2 * (psiwall - psi0[1, i]) / (dy**2)
3
4 # Top moving lid (j = -1 or ny - 1)
5 omega0[-1, i] = 2 * (psiwall - psi0[-2, i]) / (dy**2) - 2 * Ut / dy
6
7 # Left stationary wall (i = 0)
8 omega0[j, 0] = 2 * (psiwall - psi0[j, 1]) / (dx**2)
9
10 # Right stationary wall (i = -1 or nx - 1)
11 omega0[j, -1] = 2 * (psiwall - psi0[j, -2]) / (dx**2)

```

Listing 1: Numerical update of wall vorticity boundary conditions in Python

This demonstrates complete consistency between the mathematical theory ($\nabla^2\psi = -\omega$) and the discrete Python implementation.